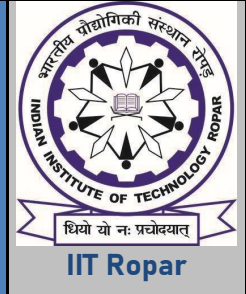


DEPARTMENT OF MECHANICAL ENGINEERING

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CP 302 CAPSTONE PROJECT MID SEMESTER REPORT

MICRO JET ENGINE

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2. Abstract

This report details the design, analysis, and preliminary fabrication of a hybrid electric-combustion micro-jet engine. The primary objective is to develop a turbine-less engine where an electric motor drives the compressor, allowing the high-enthalpy exhaust gas to expand fully through the nozzle to maximize thrust. The project encompasses a comprehensive cycle analysis based on the Brayton cycle to establish the theoretical performance parameters of the engine. Key design variables such as compressor pressure ratio, combustor exit temperature, and flight conditions are considered to predict thrust, fuel efficiency, and specific impulse. The design and methodology section outlines the iterative development of the centrifugal compressor, the annular combustion chamber, the fuel injection system, and the exhaust nozzle.

Design parameters were finalized in the DEP project; this semester we will carry out fabrication and iterative testing. Initial manufacturing has been carried out using 3D printing for rapid prototyping and testing, with plans for fabricating the final components from metal. The report also outlines the experimental setup for measuring the compressor's performance characteristics and the test plan for the complete engine assembly. Preliminary testing of the 3D-printed prototype has provided valuable insights into airflow, sealing, and component integration, which will inform future design iterations.

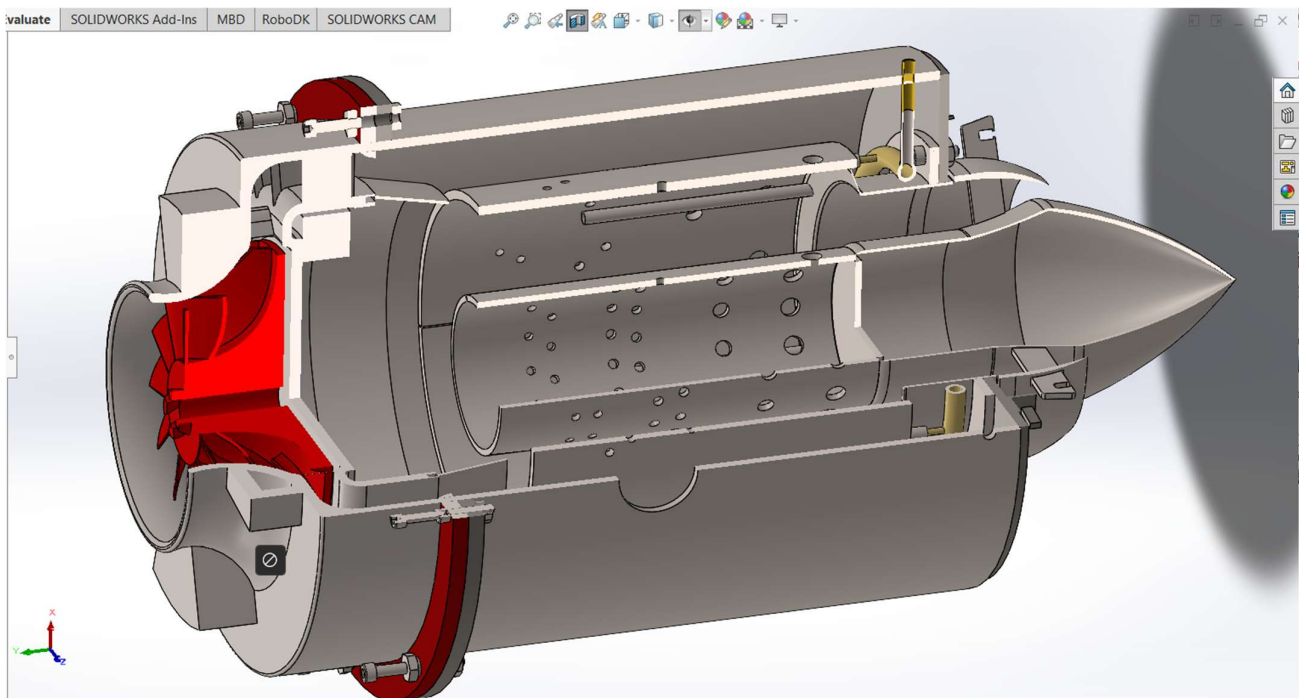


Figure 1 Quarter Sectional View of Microjet Engine

3. Introduction

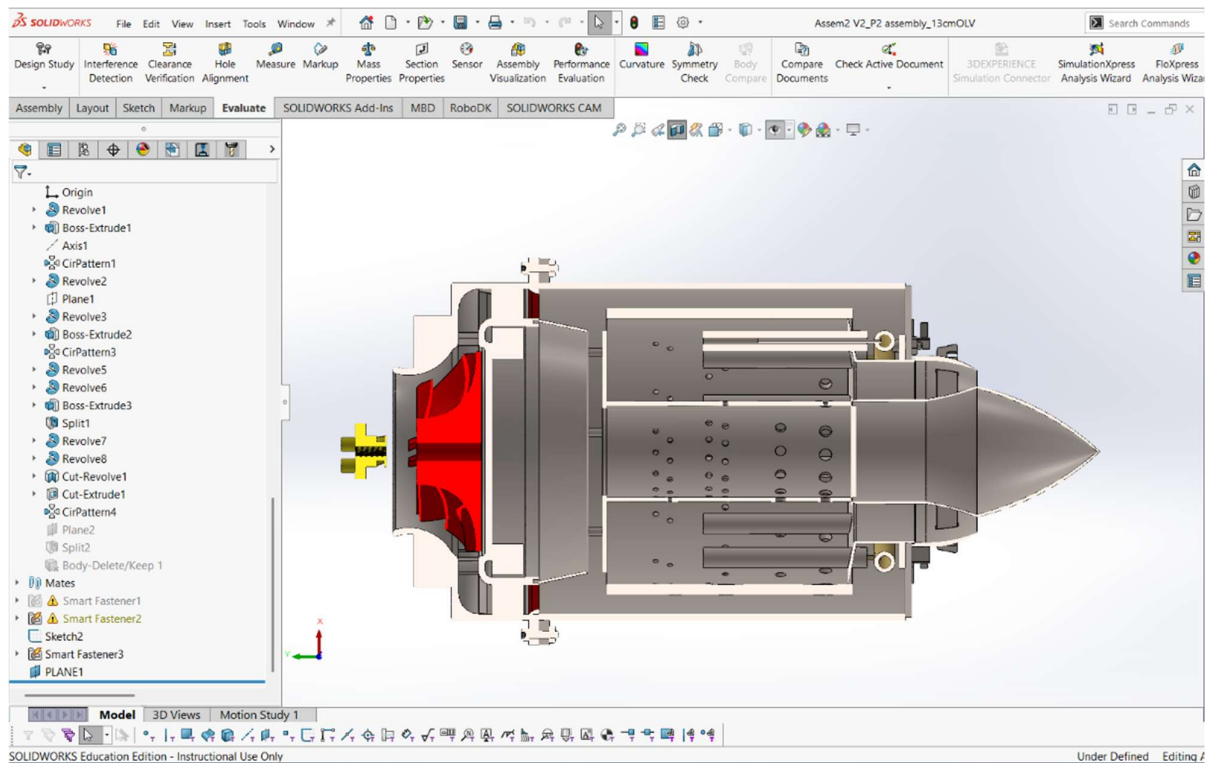


Figure 2 Half Sectional View of Micro Jet Engine

This report presents a focused thermodynamic and performance analysis of a hybrid electric–combustion micro–jet engine in which the conventional turbine is removed and the compressor is driven by an electric motor. Motivated by the need for compact, efficient propulsion for small airframes and UAVs, the study develops stage-by-stage relations for stagnation and static state variables and derives closed-form expressions for key performance metrics: exit velocity, specific thrust, fuel–air ratio, specific impulse, thermal efficiency, propulsive efficiency and overall system efficiency. Idealized assumptions (steady one-dimensional flow, isentropic compressor and nozzle, negligible combustor pressure losses unless stated) are used to isolate the thermodynamic effects of removing the turbine and to clearly quantify the electrical power required to drive the compressor.

The methodology follows the flow path freestream → compressor → combustor → nozzle and couples the compressor work (electrical input) with the combustor energy balance to compute the fuel requirement and nozzle exhaust conditions. Equations are presented in a form suitable for parametric studies (compressor pressure ratio, turbine-inlet temperature specification, motor and combustor efficiencies, flight Mach number), enabling sensitivity analysis and design tradeoffs. Results and relations herein are intended to guide component sizing, inform control and power-management decisions for the hybrid architecture, and provide the theoretical basis for the subsequent fabrication and iterative testing planned in this semester's continuation of the DEP project.

4. Background / Literature Review:

4.1 Preliminary cycle Analysis: Brayton Cycle



Ideal Brayton Cycle
T-s diagram

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Ideal Brayton Cycle
p-V diagram

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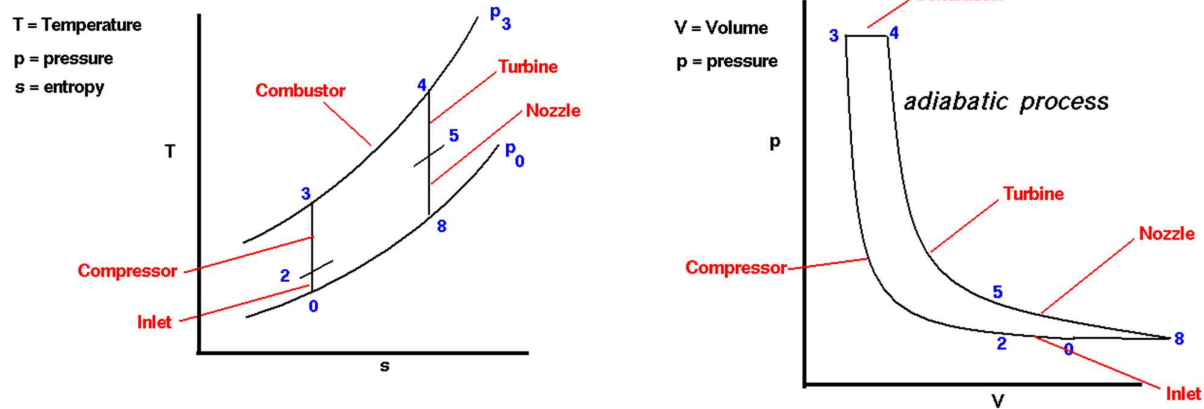


Figure 3 TS & PV diagrams of a Brayton cycle (Ideal)

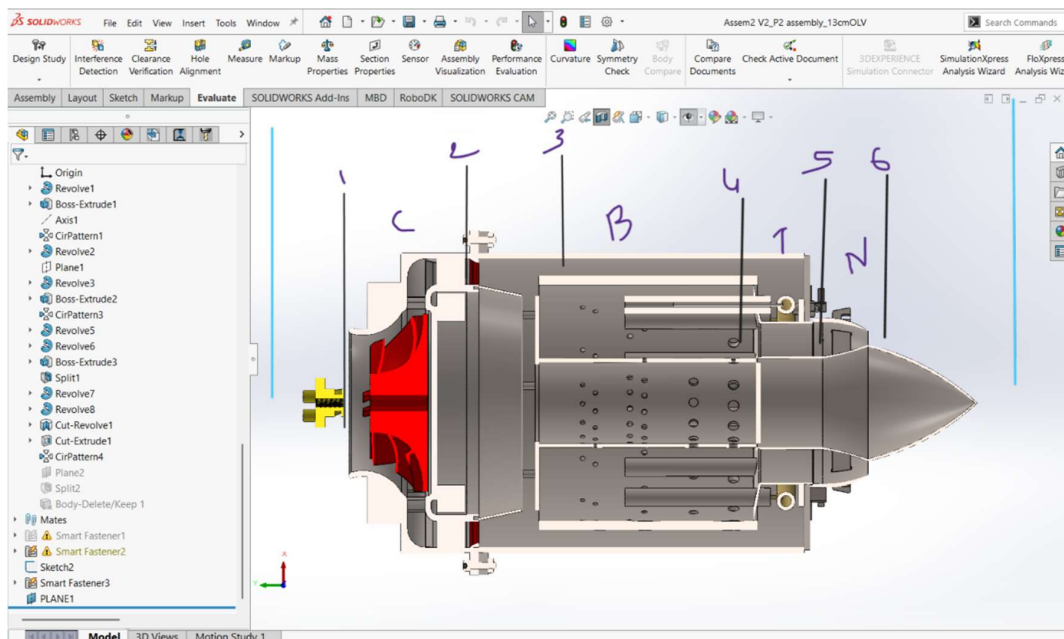


Figure 4 Notation and station numbering

In this section we will perform “ideal cycle analysis”, which is a method for expressing the thrust and thermal efficiency of engines in terms of *useful design variables*. Our objective is to express thrust, T , and thermal efficiency, h (or alternatively l) as functions of 1) typical design limiters, 2) flight conditions, and 3) design choices so that we can analyse the performance of various engines.

The expressions will allow us to define a particular mission and then determine the optimum component characteristics (e.g. compressor, combustor, turbine) for an engine for a given mission. Note that ideal cycle analysis addresses only the thermodynamics of the airflow within the engine. It does not describe the details of the components (the blading, the rotational speed, etc.), but only the results the various components produce (e.g. pressure ratios, temperature ratios).

Stagnation temperature (T_t) : The temperature a gas would reach if it were brought to a complete stop from a given velocity in a frictionless, adiabatic (isentropic) process.

It represents the sum of the **static temperature (T)**, which is due to the random thermal motion of molecules, and the **dynamic temperature**, which is due to the directed kinetic energy of the fluid flow. The stagnation temperature is the highest temperature that can be achieved in a flow and occurs at stagnation points on an object, such as the nose of an aircraft. For an ideal gas, it remains constant across a shock wave.

Derivation of the Stagnation Temperature Relation

The relationship between stagnation temperature (T_t), static temperature (T), and the ratio of specific heats (γ) is derived from the steady-flow energy equation.

1. **The Energy Equation:** The total energy of a unit mass of gas is the sum of its internal energy (represented by static enthalpy, h) and its kinetic energy. The total enthalpy, also known as stagnation enthalpy (h_t), is constant.

- $h_t = h + \frac{v^2}{2}$ For a calorically perfect gas, enthalpy is $h = c_p T$.
- $c_p T_t = c_p T + \frac{v^2}{2}$

2. **Mach Number:** Divide the equation by $c_p T$:

- $\frac{T_t}{T} = 1 + \frac{v^2}{2c_p T}$
- We know that the Mach number is $M = v/a$
- The speed of sound is $a = \sqrt{\gamma RT}$ (so $a^2 = \gamma RT$). This gives us $v^2 = M^2 \gamma RT$.

3. **Substitute and Simplify:** Substitute the expression for v^2 into the equation:

- $\frac{T_t}{T} = 1 + \frac{M^2 \gamma RT}{2c_p T}$
- $\frac{T_t}{T} = 1 + \frac{M^2 \gamma R}{2c_p}$: we know $c_p = \frac{\gamma R}{\gamma - 1}$.
- $\frac{T_t}{T} = 1 + \frac{M^2 \gamma R}{2 \left(\frac{\gamma R}{\gamma - 1} \right)}$ Cancel out the γR terms:
- $\frac{T_t}{T} = 1 + \frac{M^2 (\gamma - 1)}{2}$

This gives the final relation: $T_t = T \left(1 + \frac{\gamma-1}{2} M^2 \right)$

Similarly, $T_t = T \left(1 + \frac{\gamma-1}{2} M^2 \right)$, $P_t = P \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}}$

4.2 COMPRESSOR:

Before entering the Compressor

$$P_{1i} = 1.01 \times 10^5 \text{ Pa } T_{1i} = 300 \text{ K } \rho_{1i} = 1.204 \text{ kg/m}^3 V_{1i} = 0 \text{ m/s}$$

At 10,000 RPM

$$\dot{m} = 0.0137 \text{ kg/s } \frac{P_{1o}}{P_{1i}} = 1.0132 \Rightarrow P_{1o} = 1.01 \times 1.0132 \times 10^5 \text{ Pa } P_{1o} = 1.0233 \times 10^5 \text{ Pa}$$

General Equations

$$V_{1o} = \frac{\dot{m}}{\rho_{1o} A_1} \rho_{1o} = f(P_{1o}, T_{1o}) \Rightarrow \rho_{1o} = \frac{P_{1o}}{RT_{1o}} T_{1o} = T_{1i} \left[1 + \frac{1}{\eta} \left(\left(\frac{P_{1o}}{P_{1i}} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right) \right] R = 287 \text{ J/kgK}$$

Calculation with Efficiency

$$\text{If } \eta = 0.85, T_{1o} = 301.32 \text{ K}, \rho_{1o} = 1.1833 \text{ kg/m}^3 V_{1o} = \frac{0.0137}{1.1833 \times 0.002049} \Rightarrow V_{1o} = 5.65 \text{ m/s}$$

Just after exiting the Compressor vanes (Before entering the diffuser)

$$P_{1o} = 1.0233 \times 10^5 \text{ Pa } T_{1o} = 301.32 \text{ K } \rho_{1o} = 1.1833 \text{ kg/m}^3 V_{1o} = 5.65 \text{ m/s } \dot{m} = 0.0137 \text{ kg/s}$$

$$\dot{m} = \rho_{2i} A_2 V_{2i}$$

$$\rho_{2i} = \frac{P_{2i}}{RT_{2i}}$$

Stagnation temperature should be constant for adiabatic process:

$$T_0 = T + \frac{V^2}{2c_p} = \text{Constant}$$

$$301.32 + \frac{(5.65)^2}{2(1.005)} = T_{2i} + \frac{V_{2i}^2}{2(1.005)}$$

$$T_{2i} = 317.201 - \frac{V_{2i}^2}{1.01}$$

$$\dot{m} = \frac{P_{2i}}{R \left(317.201 - \frac{V_{2i}^2}{1.01} \right)} A_2 \cdot V_{2i}$$

It's noted that $\dot{m}$ is known, but V_{2i} is non-linearly related to $\dot{m}$. We also don't know P_{2i} . We must measure it practically.

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For example, we have measured $P_{2i} = 1.05 \times 10^5$ Pa using a water column, then:

$$V_{2i} = 2.75 \text{ m/s}$$

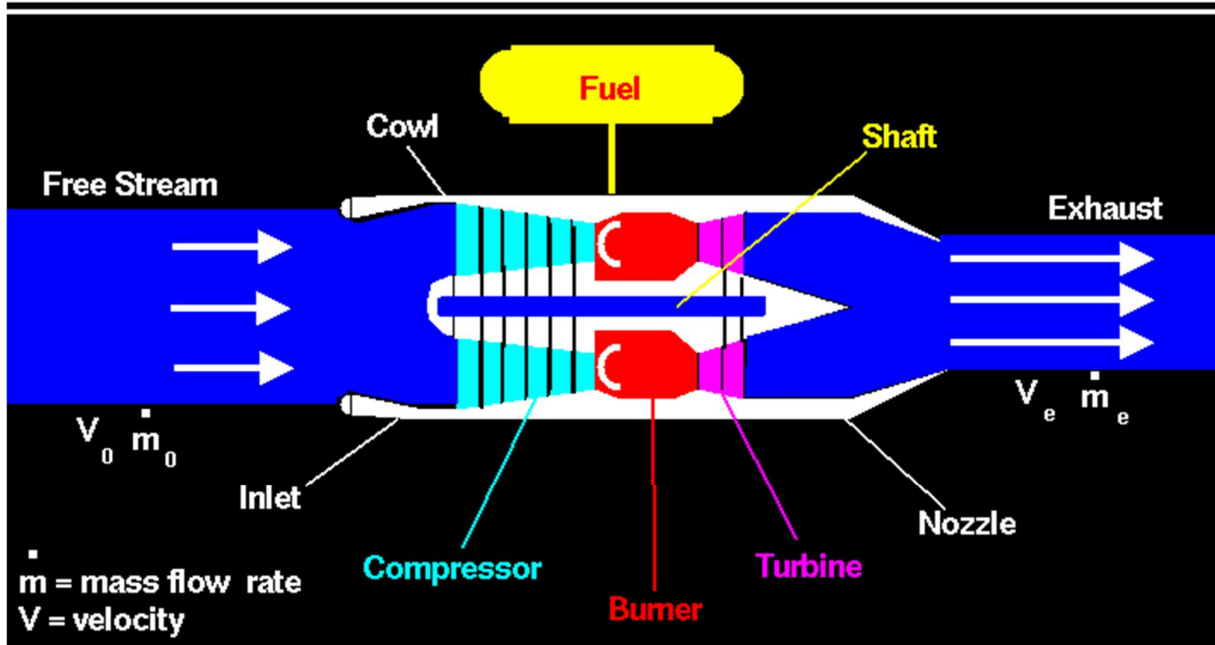
$$T_{2i} = 301.332 \text{ K}$$

$$\rho_{2i} = 1.2141 \text{ kg/m}^3$$



Turbojet Thrust

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$$\text{Thrust} = F = \dot{m}_e V_e - \dot{m}_0 V_0$$

4.3 COMBUSTION CHAMBER:

Definitions / symbols

γ — ratio of specific heats (air)

c_p — specific heat at constant pressure (air)

R — gas constant (air)

T_0, p_0, M_0, u_0 — ambient/static temperature, pressure, flight Mach and flight speed

$$(u_0 = M_0 \sqrt{\gamma R T_0})$$

T_t — stagnation (total) temperature, p_t — stagnation (total) pressure (subscripts follow stations)

$\dot{m}_a$ — mass flow of air (kg/s)

f — fuel/air mass ratio ($\dot{m}_f / \dot{m}_a$)

LHV — lower heating value of fuel (J/kg)

η_b — combustor (combustion) efficiency (fraction)

$\pi_c \equiv p_{t3}/p_{t2}$ — compressor total-pressure ratio (design variable)

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Assume **ideal isentropic compressor and nozzle**, negligible pressure losses in combustor ($p_{t4} = p_{t3}$), ideal fully-expanded nozzle ($p_e = p_0$) unless otherwise stated.

Combustor / burner (3 → 4)

Assume negligible pressure loss: $p_{t4} = p_{t3}$. Energy balance (steady flow, per kg air):

$$f \text{ LHV } \eta_b = c_p (T_{t4} - T_{t3}) \quad \Rightarrow \quad f = \frac{c_p (T_{t4} - T_{t3})}{\eta_b \text{ LHV}}.$$

Equivalently specify target turbine-inlet temperature (here turbine absent) or combustor exit T_{t4} directly and compute f from above.

Fuel	LHV (MJ/kg)	LHV (BTU/lb)
Natural Gas (Methane)	50.0	21500
Diesel	43.4	18700
Gasoline	44.0	18900
Kerosene	43.0	18500
Hydrogen	120.0	51600
Propane	46.35	19900
Coal (Anthracite)	32.5	14000

Material	Maximum Continuous Temperature (°C)	Maximum Intermittent Temperature (°C)	Melting Point (°C)	Notes
Mild Steel	400-650	750-900	1450-1520	Strength drops significantly at 400-650°C
Aluminum (General)	150-300	300-350	660	Most alloys lose strength above 200°C
Aluminum 6061	200-250	300	660	Heat-treated alloy, good up to 250°C
Stainless Steel 304	870	925	1400-1450	Excellent high-temp performance
Stainless Steel 316	870	900	1371-1399	Similar to 304, slightly lower melting point

Nozzle (4 → e exit)

Stagnation at nozzle entry: T_{t4} , $p_{t4} = p_{t3}$. For isentropic nozzle expanding to ambient ($p_e = p_0$):

$$\frac{T_e}{T_{t4}} = \left(\frac{p_e}{p_{t4}}\right)^{(\gamma-1)/\gamma} = \left(\frac{p_0}{p_{t4}}\right)^{(\gamma-1)/\gamma}.$$

Exit static temperature:

$$T_e = T_{t4} \left(\frac{p_0}{p_{t4}}\right)^{(\gamma-1)/\gamma}.$$

Exit velocity (from energy equation):

$$u_e = \sqrt{2 c_p (T_{t4} - T_e)}.$$

Exit Mach (if needed):

$$M_e = \sqrt{\frac{2}{\gamma - 1} \left(\frac{T_{t4}}{T_e} - 1\right)}.$$

4.4 Thrust and performance relations

Thrust (steady 1D, per unit air mass flow)

Assuming fully expanded nozzle ($p_e = p_0$ so pressure thrust term zero):

$$\frac{F}{\dot{m}_a} = (1 + f) u_e - u_0$$

(If we keep the pressure term: momentum Thrust + $(p_e - p_0)A_e$.)

Specific thrust (per kg air)

Same as above: $F/\dot{m}_a = (1 + f)u_e - u_0$.

Fuel/air ratio from combustor specification

$$f = \frac{c_p (T_{t4} - T_{t3})}{\eta_b LHV}$$

Exit velocity expressed via design variables

Substituting, $T_{t3} = T_{t2}\pi_c^{(\gamma-1)/\gamma}$ and $T_{t2} = T_{t0}$:

$$T_{t4} \text{ (given)} \Rightarrow u_e = \sqrt{2c_p \left(T_{t4} - T_{t4} \left(\frac{p_0}{p_{t4}} \right)^{(\gamma-1)/\gamma} \right)}$$

with $p_{t4} = p_{t3} = p_{t2}\pi_c$ and $p_{t2} = p_{t0}$ from freestream, so $p_{t4} = p_{t0}\pi_c$. Thus u_e can be written fully in terms of T_{t4} , T_{t0} , π_c , p_0 .

More compact when you define:

$$q_t \equiv \frac{T_{t4}}{T_0}, \Rightarrow u_e = \sqrt{2c_p T_{t4} (1 - (p_0/p_{t0}\pi_c)^{(\gamma-1)/\gamma})}.$$

Specific impulse (ideal)

$$I_{sp} = \frac{F/\dot{m}_a}{\dot{m}_f/\dot{m}_a g_0} = \frac{(1 + f)u_e - u_0}{f g_0}.$$

For small f, approx. $I_{sp} \approx \frac{u_e - u_0}{f g_0}$.

Thermal efficiency (engine) — include compressor work penalty

Fuel energy input per kg air: $f LHV$. Useful *propulsive* power per kg air:

$$P_{prop} = (1 + f) \frac{u_e^2 - u_0^2}{2}.$$

If compressor shaft work W_c is provided by the fuel (i.e. internal fuel power pays compressor), net useful power becomes $P_{net} = P_{prop} - W_c$. Thus ideal **thermal efficiency** (net propulsive power divided by fuel chemical power) is:

$$\eta_{thermal} = \frac{(1 + f) \frac{1}{2} (u_e^2 - u_0^2) - W_c}{f LHV}.$$

With small f and neglect of fuel mass in kinetic term, numerator $\approx 0.5(u_e^2 - u_0^2) - W_c$.

If the compressor work is provided by an external shaft (not from the engine fuel), drop W_c term.

Propulsive efficiency

Standard expression (for small f):

$$\eta_{\text{prop}} = \frac{\text{propulsive power}}{\text{kinetic power increase}} = \frac{(1+f)u_0(u_e - u_0)}{(1+f) \frac{1}{2} (u_e^2 - u_0^2)} = \frac{2u_0}{u_e + u_0}$$

(This formula is independent of f if the exhaust mass $\approx$ intake mass.)

Overall efficiency

Overall (engine) efficiency = propulsive power / fuel power:

$$\eta_{\text{overall}} = \eta_{\text{thermal}} \times \eta_{\text{prop}}$$

Or from definitions: $\eta_{\text{overall}} = \frac{(1+f) \frac{1}{2} (u_e^2 - u_0^2) - W_c}{f \text{ LHV}} \times \frac{2u_0}{u_e + u_0}$, the expanded form

1. Removing the turbine means **compressor work must be supplied externally** from separate power. If the compressor shaft power comes from the engine fuel, include W_c in the fuel energy budget (as above). For external power the derivation for efficiency is done in later section.
2. No diffuser $\Rightarrow$ stagnation recovery at compressor inlet is just freestream stagnation (no additional pd or td gains).
3. If combustor pressure losses exist, set $p_{t4} = p_{t3} \cdot \beta_b$ ($\beta_b < 1$), and for imperfect expansion negligible term pressure thrust term $(p_e - p_0)A_e$ will be included in further iteration of calculations.

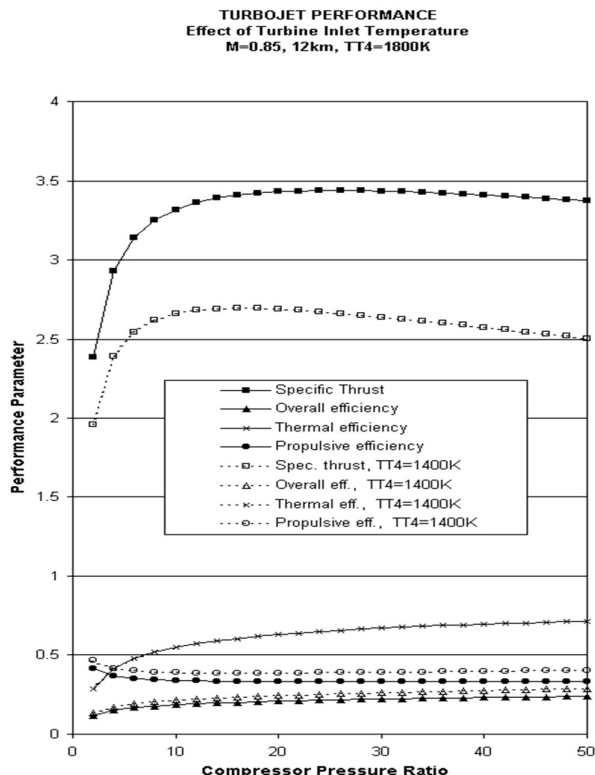


Figure 5 Performance of an ideal turbojet engine as a function of compressor pressure ratio and turbine inlet temperature

EFFICIENCY CALCULATION FOR ELECTRIC MICRO JET ENGINE:

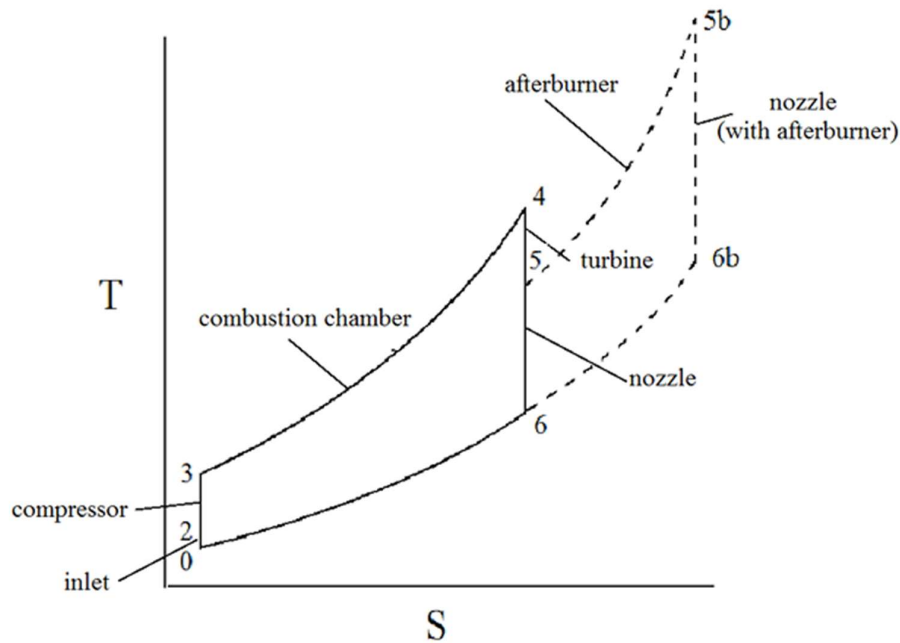


Figure 4: TS diagram of the ideal Brayton Cycle for Turbojet

1. Conventional Turbojet Thermal Efficiency

The performance of a standard gas turbine (turbojet) engine is typically analyzed using the **Brayton cycle**. The thermal efficiency (η_{th}) defines how effectively the heat energy from the fuel is converted into useful net work.

For an ideal Brayton cycle:

$$\eta_{th} = \frac{W_{net}}{Q_{in}}$$

where:

- **Net Work:**

$$W_{net} = (h_3 - h_4) - (h_2 - h_1)$$

(the turbine work output minus the compressor work input)

- **Heat Input:**

$$Q_{in} = h_3 - h_2$$

(the enthalpy rise across the combustor)

Here, h_1, h_2, h_3, h_4 denote the specific enthalpies at the compressor inlet, compressor outlet, turbine inlet, and turbine outlet, respectively.

2. The Hybrid Electric-Combustion Concept

In the proposed engine, the turbine is eliminated and the compressor is instead driven by an **electric motor**. This changes the energy balance:

- Compressor work requirement is supplied electrically, not by turbine shaft power.
- The high-enthalpy gas leaving the combustor expands completely through the nozzle, maximizing jet thrust.

Thus, the useful output is **jet kinetic energy** → **thrust power**, while the inputs are **fuel heat release + electrical motor power**.

3. Thrust Power as Useful Work

For a jet engine, the useful work output is the **propulsive power** it delivers to the aircraft. This is known as thrust power:

$$P_{thrust} = T \cdot V_a$$

where:

- T = net thrust,
- V_a = flight velocity of the aircraft.

This represents the rate of useful work done by the engine to propel the aircraft forward.

4. Efficiency Formulations for the Hybrid Engine

4.1. Thermal Efficiency

Thermal efficiency considers only the chemical (fuel) energy input and measures how effectively combustion energy is converted into thrust power.

$$\eta_{th} = \frac{P_{thrust} - W_c}{\dot{Q}_{in}}$$

Substituting:

$$\eta_{th} = \frac{T \cdot V_a - W_c}{\dot{m}_f Q_{HV}}$$

where:

- $\dot{m}_f$ = fuel mass flow rate,
- Q_{HV} = heating value of the fuel (kJ/kg).
- W_c = Work done by turbine

$P_{thrust} = h_3 - h_4$, h_3 contains all the energy imparted by work from compressor and heat input from addition of fuel, h_4 is the heat lost to the surrounding, $h_3 - h_4$ is the work done in iso entropic expansion, $h_3 - h_4 - W_c$ contains only the fraction of work

extracted from heat , so the ratio $(h_3 - h_4 - W_c)/Q$ is the fraction of heat converted to work , (Thermal Efficiency)

4.2. Overall Energy Efficiency

The **overall efficiency** accounts for both chemical and electrical energy inputs:

$$\eta_{overall} = \frac{P_{thrust}}{Q_{in} + P_{motor}}$$

Substituting:

$$\eta_{overall} = \frac{T \cdot V_a}{\dot{m}_f Q_{HV} + P_{motor}}$$

where:

- P_{motor} = electrical power supplied to drive the compressor.

4.3. Note on Motor Power

The instantaneous **rotational kinetic energy** of the compressor is:

$$E_{rotor} = \frac{1}{2} I \omega^2$$

However, for steady-state operation, the relevant input is **power**, not stored energy. The motor must continuously supply the compression work:

$$P_{motor} = \dot{m}_{air} (h_2 - h_1)$$

where:

- $\dot{m}_{air}$ = air mass flow rate through the compressor,
- $(h_2 - h_1)$ = enthalpy rise across the compressor.

4.4. Summary of Efficiency Equations

- **Thermal Efficiency (fuel only):**

$$\eta_{th} = \frac{T \cdot V_a - W_c}{\dot{m}_f Q_{HV}}$$

- **Overall Efficiency (fuel + motor):**

$$\eta_{overall} = \frac{T \cdot V_a}{\dot{m}_f Q_{HV} + P_{motor}}$$

- **Droplet Sauter mean diameter (d_{32}):** For liquid fuels, $d_{32} \lesssim 20 \mu\text{m}$ to ensure vaporization within chamber residence time.

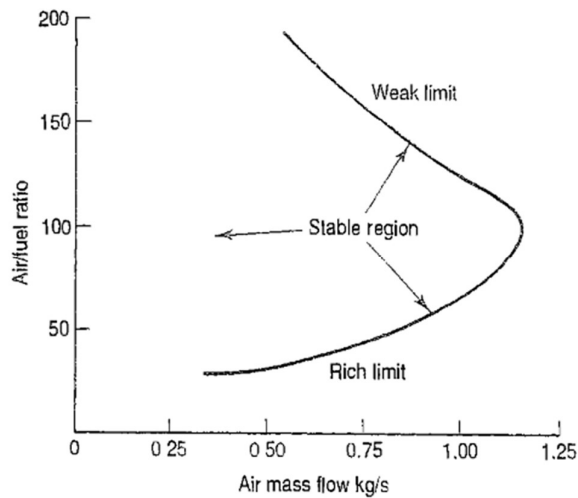


FIG. 6.7 Stability loop

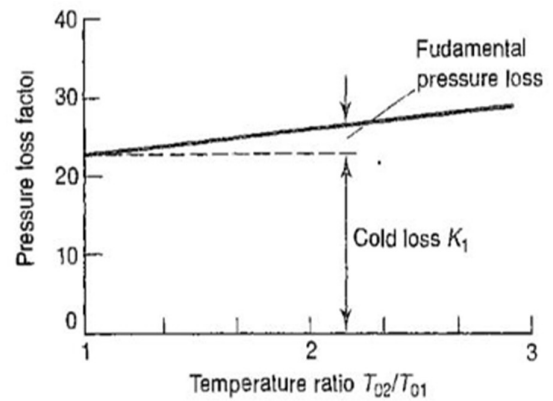


FIG. 6.5 Variation of pressure loss factor

5.3 Governing Parameters

- Arrhenius reaction rate:

$$k = A e^{-E_a/(RT)}$$

- Minimum ignition energy (approx.):

$$E_{min} = \frac{4}{3} \pi R_{cr}^3 \rho c_p (T_{ad} - T_u)$$

- Critical ignition radius / thermal thickness:

$$R_{cr} = \delta_L \approx \frac{S_L}{\alpha_u}$$

- Flame propagation with stretch:

$$U = S_L \left(1 - \frac{\kappa Le}{R_f} \right) \quad \text{where } \kappa = \frac{1}{R} \frac{dR}{dt}$$

- Heat release rate:

$$\dot{q} = \dot{m}_f \cdot LHV \cdot \eta_c$$

- Dam Köhler number (practical form):

$$Da = \frac{\tau_{res}}{\tau_{chem}} \approx \tau_{res} \cdot A[R]^n e^{-E_a/(RT)}$$

- Pressure loss coefficient (empirical):

$$\omega_{cc} = K_1 + K_2 \left(\frac{T_3}{T_2} - 1 \right)$$

5.4 Governing conservation equations (PDE set)

- **Mass:**

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- **Momentum:**

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho \mathbf{g}$$

- **Energy:**

$$\rho c_p \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) = \nabla \cdot (k \nabla T) + \dot{q}_{react} + \dot{q}_{visc}$$

- **Species (i):**

$$\frac{\partial (\rho Y_i)}{\partial t} + \nabla \cdot (\rho Y_i \mathbf{u}) = \nabla \cdot (\rho D_i \nabla Y_i) + \omega_i$$

Simultaneously solving these (RANS/LES + detailed or reduced kinetics) and checking coupled constraints.

5.5 Key dimensionless groups (two-line each)

- **Reynolds (Re) = $\rho V D / \mu$:** Determines turbulence level; higher Re $\rightarrow$ turbulent mixing, shorter τ_{res} locally.
- **Prandtl (Pr) = $\mu c_p / k$:** Relates momentum to thermal diffusion; affects boundary layer and cooling design.
- **Lewis (Le) = α / D :** Thermal vs mass diffusivity ratio; controls preferential diffusion effects.
- **Zeldovich (Ze):** Activation-energy sensitivity indicator for ignition and branch reactions.
- **Frank–Kamenetskii (δ):** Thermal explosion parameter — useful when evaluating quench/thermal-runaway risk.

5.6 Compact design constraints & target ranges

Constraint	Formula / Test	Typical target (micro-annular)
Residence vs chemistry	$\tau_{res} \geq \tau_{chem}$ ($Da \gtrsim 1$)	$\tau_{res} \approx 1\text{--}20$ ms
Equivalence ratio	$\phi \approx 0.8 - 1.2$	0.9 ± 0.2
Axial velocity vs flame speed	$V_{axial} < S_L$ (or ensure recirc)	$V_{axial} \lesssim 0.5\text{--}2$ m/s (scale dep.)
Swirl number	$S \gtrsim 0.6$ for CRZ	$S = 0.6\text{--}1.2$
Pressure loss	$\Delta P/P \lesssim 0.03\text{--}0.07$	3–7% of compressor outlet
Material temp limit	$T_3 \leq T_{max_material}$	1100–1700 K (material dep.)
Droplet size (liquid fuel)	$d_{32} \lesssim 20$ μm	≤ 20 μm

5.7 Practical design checklist

- Ensure τ_{res} ($V/\tilde{V}$) and recirculation topology produce $Da \geq 1$.
- Target ϕ in the 0.8–1.2 band; provide local mixing maps at recirculation zone.
- Design swirlers and primary zone volume fraction ($\approx 20\text{--}30\%$) to anchor flame in CRZ.
- Limit ΔP to 3–7% of compressor outlet; quantify with CFD losses.
- Specify S_L (mixture) and ensure local axial velocities are lower or balanced by recirculation.
- Implement film/effusion cooling and dilution rows sized to meet turbine T_{max} limits and uniformity requirements.
- For liquid fuels, design atomizer for $d_{32} \leq 20 \mu\text{m}$; verify vaporization with Lagrangian spray model.
- Define ignition system to deposit E_{ign} above E_{min} over a kernel radius $\geq R_{cr}$.

5.8 Modelling & verification steps

- **OD/ID quick checks:** Use Da , ϕ and energy balance to screen candidate geometries before CFD.
- **RANS CFD (with reduced kinetics):** Compute flow, recirculation, ΔP and mean temperature field; include transport-corrected S_L .

5.9 Conclusion

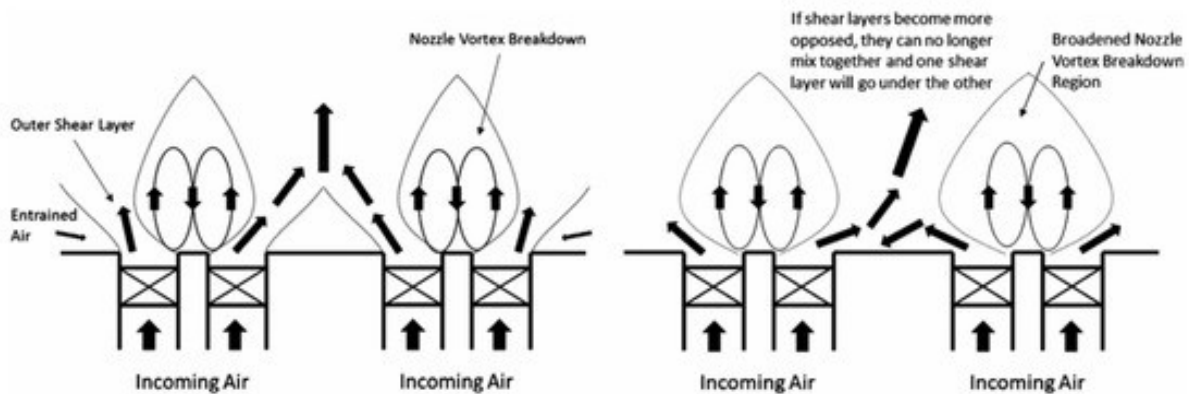


Figure 7 Flow Recirculation Mechanisms for Flame Stabilization

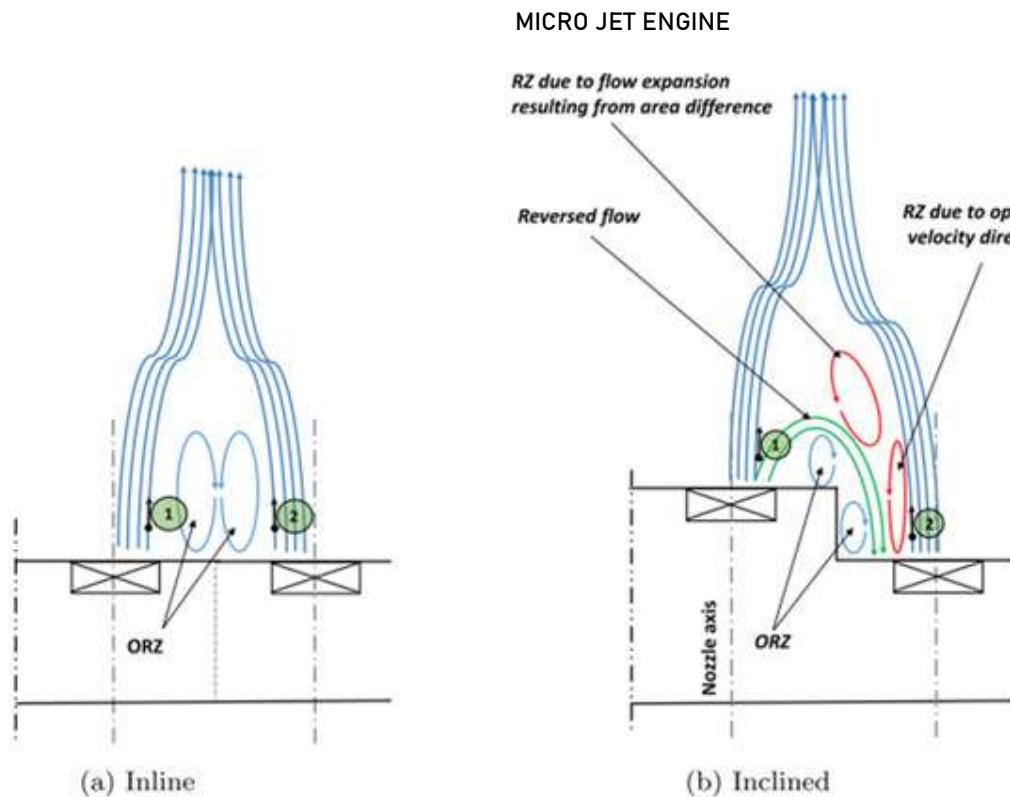


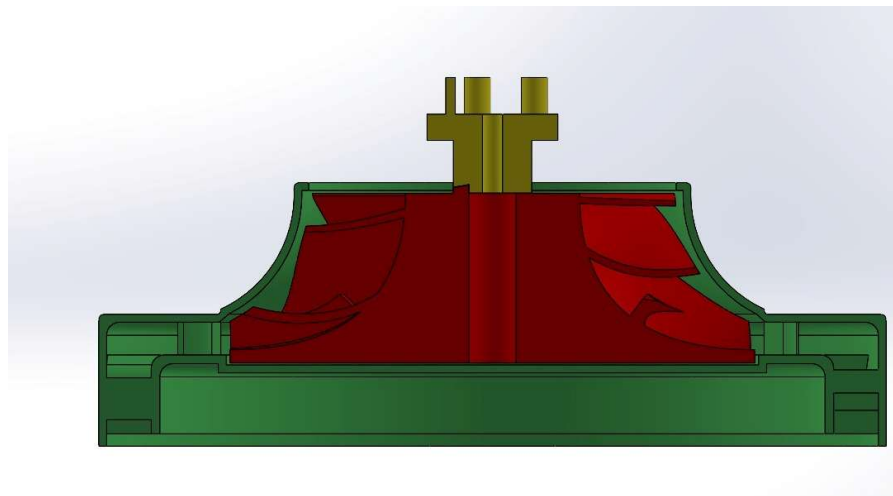
Figure 8 Vortex Breakdown as a Flame Holding Mechanism

Prioritize residence time (Da), equivalence ratio, flame speed vs axial velocity and recirculation strength — these four dominants govern flame anchoring. Apply a staged verification: analytical checks → RANS CFD → targeted LES/experiments for ignition and transient stability.

6. Design and Methodology

6.1 Design and Making of Centrifugal Compressor:

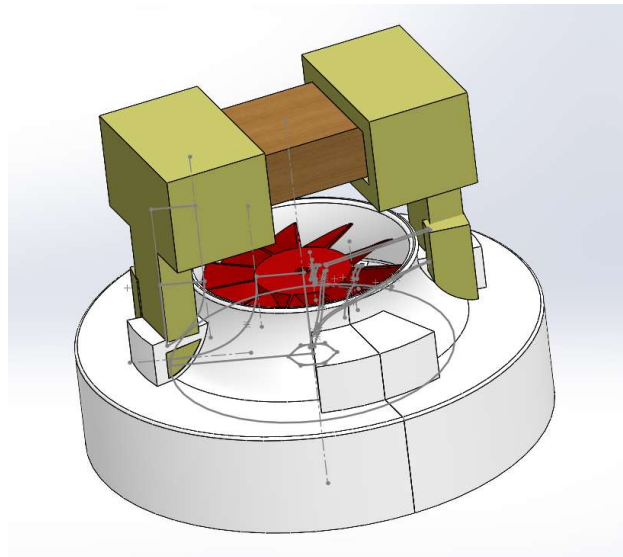
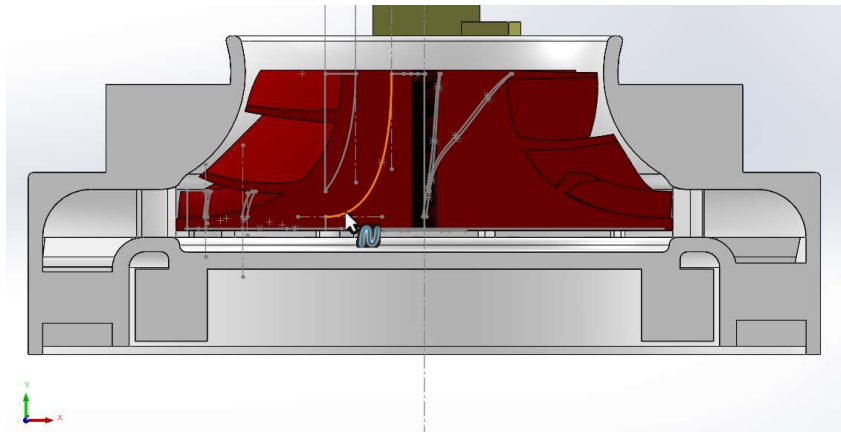
Previous Design:



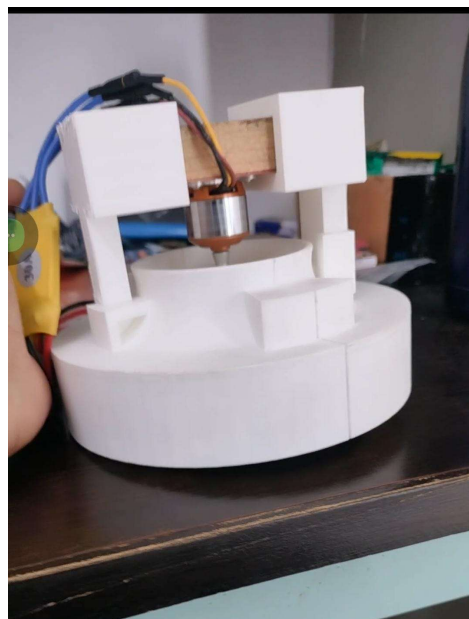
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Modified design:

This completely disassemble design allows us to modify the parts very easily.



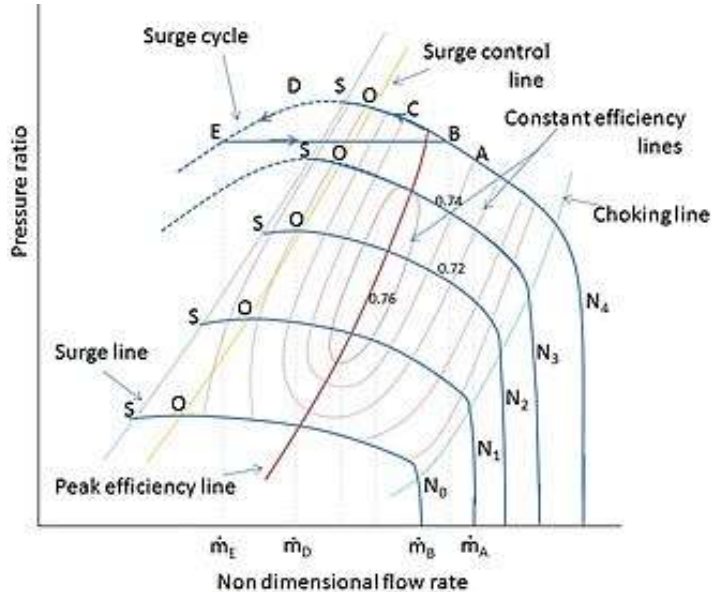
After Mounting the Motor:



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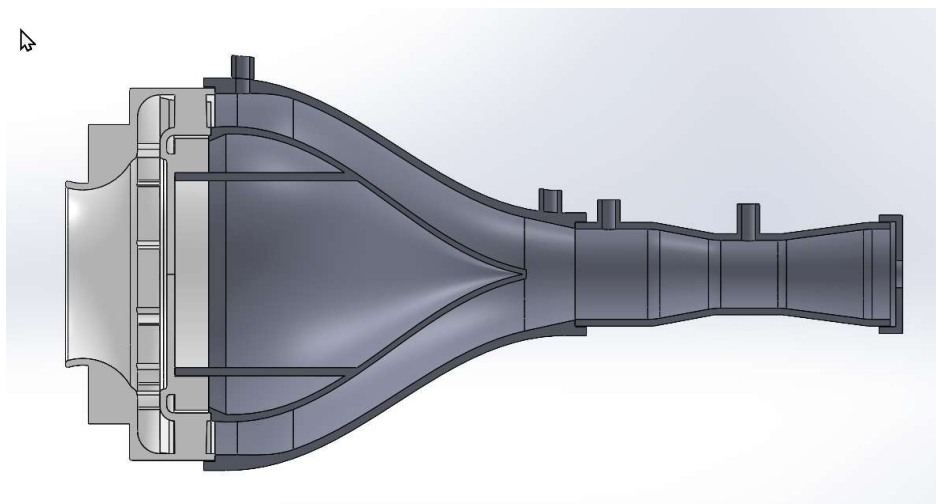
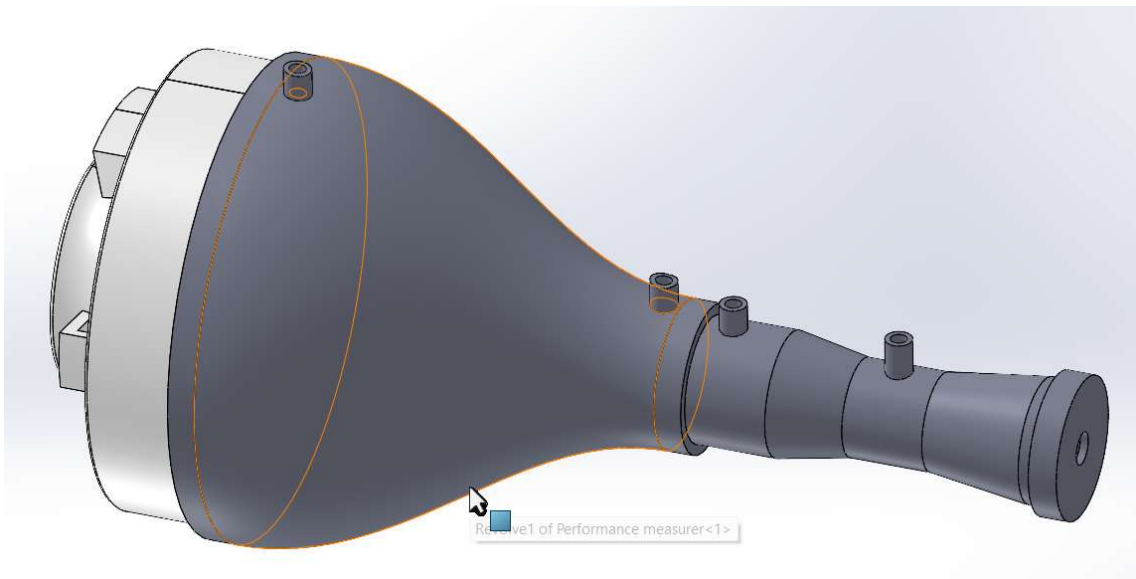
The compressor is rotating very smoothly and giving a good amount of mass flow rate.

Our next job is to plot the performance curve of the compressor for at least three different RPMs.



We have to find the surge line and avoid the mass flow rate go below the surge line.

- For finding the performance characteristics, we need to measure two things i.e., pressure at the exit of compressor and the mass flow rate.
- To measure that, we have designed an Apparatus separately:



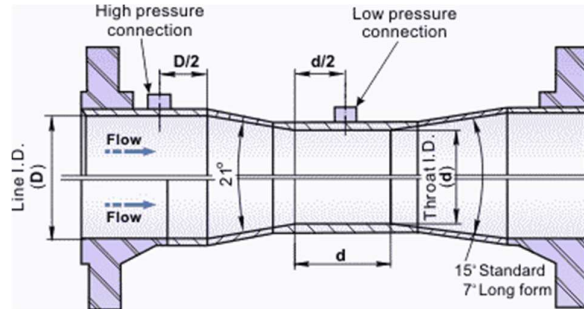
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The air flows through the large duct and the venturi meter attached to it at the right end. The first two tube extensions were to measure the static pressure at the two locations. And the remaining two on the venturi meter are to measure the mass flow rate.

Two water level tubes are attached to the larger duct to measure the static pressure and a U-shaped tube goes through the two extensions from venturi meter to measure the pressure difference between the two different cross sections.

Design of the setup:

Standard design of a venturi meter:



Mass flow rate expression:

Mass Flow Rate through an Orifice/Nozzle

$$\dot{m} = C \cdot \epsilon \cdot A_2 \cdot \sqrt{\frac{2\rho_1 \Delta p}{1 - \beta^4}}$$

Where:

- C : Discharge coefficient (typically 0.985)
- ϵ : Expansibility factor (accounts for gas compressibility)
- A_2 : Area at throat,

$$A_2 = \frac{\pi d^2}{4} \quad \text{where } d \text{ is throat diameter}$$

- ρ_1 : Upstream gas density
- $\Delta p = p_1 - p_2$: Differential pressure (inlet – throat)
- $\beta = \frac{d}{D}$: Diameter ratio (throat diameter / pipe diameter)

Expansibility Factor (ϵ)

This corrects for density change as the gas expands passing through the throat. For an ideal gas under isentropic expansion:

$$\epsilon = \left[\frac{1 - \tau^{1/\kappa}}{(1 - \beta^4)(1 - \tau)} \right]^{1/2}$$

Where:

- $\tau = \frac{p_2}{p_1}$: Pressure ratio (throat / inlet)
- $\kappa = \frac{C_p}{C_v}$: Isentropic exponent (ratio of specific heats)
- $\beta = \frac{d}{D}$: Diameter ratio (as defined above)

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- The design can be at three levels:
 - Design for turbine less engine with constant thrust/RPM:

When there is no turbine, we don't have to start the engine from a minimum thrust value. We can just drive the motor at max RPM and if we just make sure that the combustion chamber's resistance is not letting the compressor into surge, that's enough.

- Design for turbine less engine with variable thrust/RPM.

For this design, we need to make sure that for a wide range of RPMs, the compressor should not go into surge. For this design, we have to vary the RPM of motor from 100% to the minimum at each of the RPMs, we will plot the mass flow rate vs RPM and Pressure ratio vs RPM.

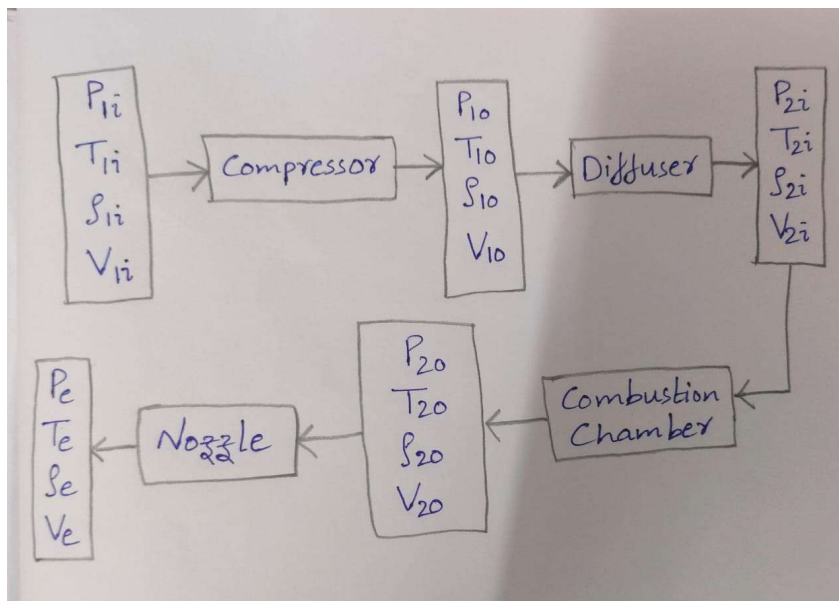
The mass flow rate and peak pressure data can help us later when examining (Testing) the combustion chamber as the velocity of the air cannot go beyond the optimal range for sustaining the flame.

- Design for engine with turbine with variable Thrust and RPM (Final Goal).

General practice to vary the mass flow rate is to use a flow control valve. Basically, the combustion chamber will act like a partially closed valve due to its intricate flow passages. And we should make sure the combustion chamber as a partially closed valve is not too closed such that the compressor is always above the surge line. And also, as close to the peak pressure point as close as possible at least for a good range of RPMs.

Calculations for the state variables:

- Nomenclature: We are dealing with compressible flow. So, there will be four variables at each stage i.e., Pressure, Temperature, Density, Velocity flow.
- In a incompressible flow, all the state variables and even gas properties are all interdependent on each other making the perfect calculation extremely difficult. So, between the ends of compressor, all gas properties are assumed to be constant as temperature doesn't change that much.
- For calculating the non-linear equations, **Newton Raphson iteration method** is being used.



$$\begin{aligned} P_{1i} &= 1.01 \times 10^5 \text{ Pa} \\ T_{1i} &= 300 \text{ K} \\ \rho_{1i} &= 1.204 \text{ kg/m}^3 \\ V_{1i} &= 0 \text{ m/s} \end{aligned}$$

→ Before entering the Compressor.

At 10,000 rpm, $\dot{m} = 0.0137 \text{ kg/s}$

$$\frac{P_{10}}{P_{1i}} = 1.0132$$

$$\Rightarrow P_{10} = 1.01 \times 1.0132 \times 10^5 \text{ Pa}$$

$$P_{10} = 1.0233 \times 10^5 \text{ Pa}$$

$$V_{10} = \frac{\dot{m}}{\rho_{10} A_1}$$

$$\rho_{10} = f(P_{10}, T_{10}) \rightarrow \rho_{10} = \frac{P_{10}}{R T_{10}}$$

$$R = 287 \text{ J/kgK}$$

$$T_{10} = T_{1i} \left[1 + \frac{1}{\eta} \left(\left(\frac{P_{10}}{P_{1i}} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right) \right]$$

If $\eta = 0.85$, $T_{10} = 301.32 \text{ K}$, $\rho_{10} = 1.1833 \text{ kg/m}^3$

$$V_{10} = \frac{0.0137}{1.1833 \times 0.002049} \Rightarrow V_{10} = 5.65 \text{ m/s}$$

$$\begin{aligned} P_{10} &= 1.0233 \times 10^5 \text{ Pa} \\ T_{10} &= 301.32 \text{ K} \\ \rho_{10} &= 1.1833 \text{ kg/m}^3 \\ V_{10} &= 5.65 \text{ m/s} \\ \dot{m} &= 0.0137 \text{ kg/s} \end{aligned}$$

→ Just after exiting the Compressor vanes. (Before entering the diffuser.)

Figure 9 State Variable Calculation at Compressor Outlet

$$\dot{m} = \rho_{2i} A_2 V_{2i}$$

$$\rho_{2i} = \frac{P_{2i}}{R T_{2i}}$$

Stagnation temperature should be constant for adiabatic process:

$$T_0 = T + \frac{V_{2i}^2}{2C_p} = \text{Constant}$$

$$301.32 + \frac{(5.65)^2}{2(1.005)} = T_{2i} + \frac{V_{2i}^2}{2(1.005)}$$

$$T_{2i} = 317.201 - \frac{V_{2i}^2}{1.01}$$

$$\dot{m} = \frac{P_{2i}}{R \left(317.201 - \frac{V_{2i}^2}{1.01} \right)} A_2 \cdot V_{2i}$$

known

V_{2i} is non-linearly related to $\dot{m}$.
We also don't know P_{2i} . We must measure practically.

★ For example, we have measured $P_{2i} = 1.05 \times 10^5 \text{ Pa}$
using water column, then, $V_{2i} = 2.75 \text{ m/s}$
 $T_{2i} = 301.332 \text{ K}$
 $\rho_{2i} = 1.2141 \text{ kg/m}^3$

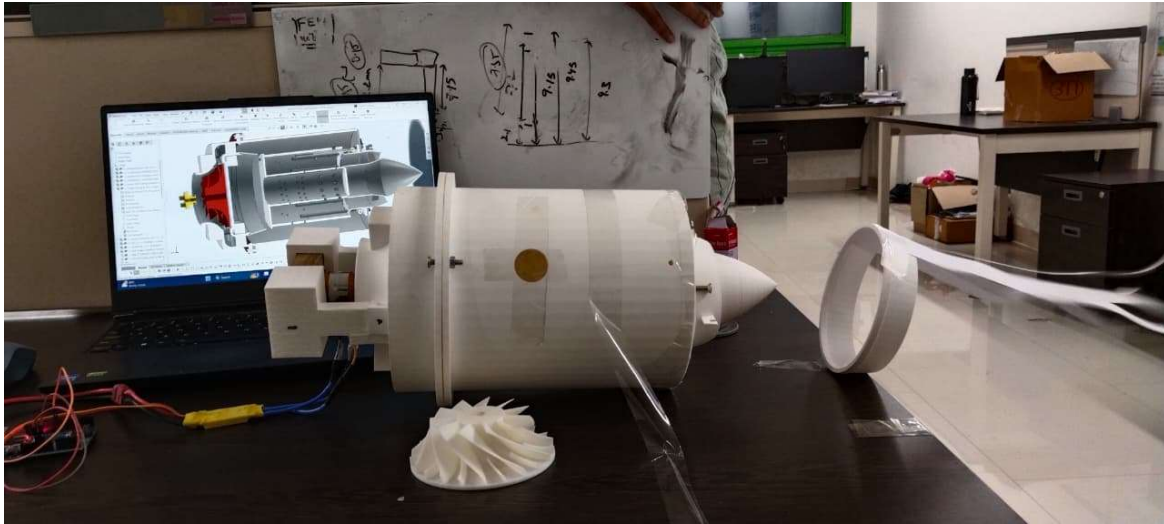
Figure 10 Iterative Calculation for Diffuser Inlet Conditions

Once we plot the performance curve of the compressor for the already designed combustion chamber as a partially closed valve, we will directly use the mass flow rate and pressure ratio relation obtained from curve fitting of those plots.

We also measure the pressure P_{2i} using the water column and use that in the calculations done above.

7. Results and Discussion

TESTING MASS FLOW RATE WITH 3D PRINTED 1:1 MODEL



There were some leakages at compressor combustion chamber joints, use of rubber gaskets can help with better sealing and also the compression is not as much as expected and velocity is high so we would improve the diverging duct design and to increase the pressure of moving fluid, in further iterations.



Figure 11 3D printed left vs Conventional (max dimensional inaccuracy 1.5mm, along length)

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Figure 12 Exhaust joint outer ring, cut with laser cutter on SS 1mm sheet metal.



Figure 13 Fuel injection setup, 11 injector tubes placed

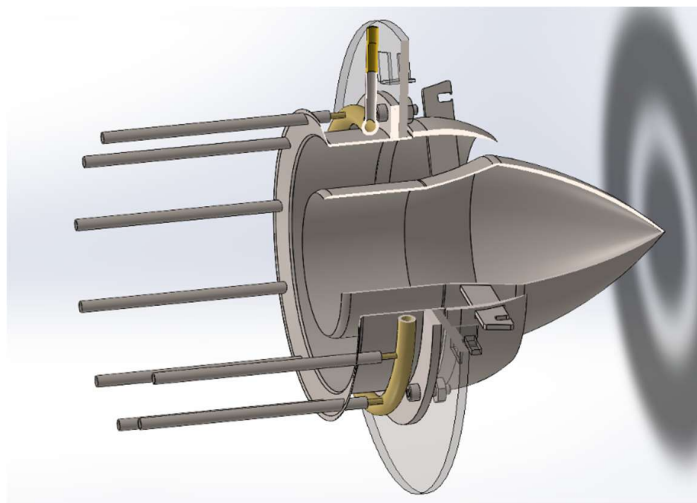
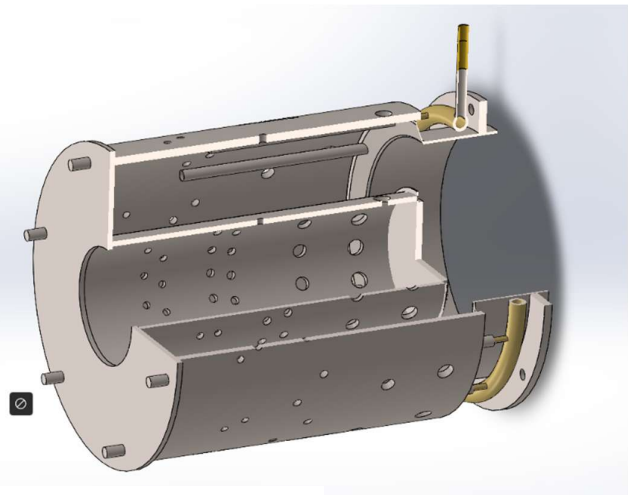


Figure 14 Combustion chamber and fuel injector, sub assembly



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Figure 15

New Impeller(left) with 1.5mm reduced clearance with compressor casing, Old Impeller (right)



Figure 16

Mass flow rate test setup (left), Outer casing, mild steel(middle); 3d printed PLA (right)



Figure 17

Acrylic sheets, mesh for test rig enclosure, outer liner (5'), sheet metal (1mm mild steel). Rubber gaskets, drill bits (HSS [2-6 mm]), nut and bolts for assembly



Figure 18 Demo for Airforce

8. Findings and Conclusion

1) Outer Casing:



The outer casing defines the external boundary for the airflow from the compressor. Compressed air flows into the space between this outer casing and outer liner. The casing directs this air into the outer liner through respective holes for combustion, while the rest is used for cooling. Cooling is essential to prevent the outer liner from melting.

2) Inner Liner:



Its perforations are designed to inject compressed air radially outward into the combustion zone. This ensures the fuel mixes properly from the inside of the annular channel.

3) Outer Liner:



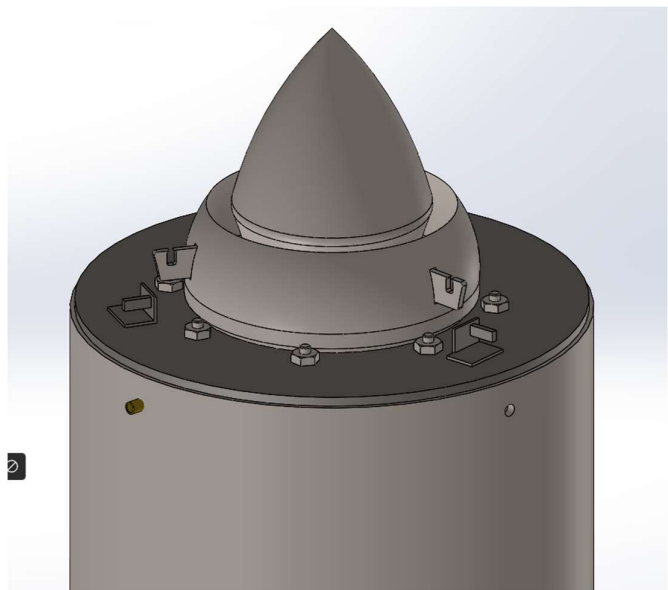
Its perforations inject compressed air radially inward into the combustion zone. This works in concert with the inner liner to create a stable, efficient, and complete burn. This airflow is critical for the liner's own survival, as it prevents the liner's walls from melting under direct flame exposure.

4) Fuel injection system:



Its job is to deliver a precise, finely vapourised spray of fuel into the combustion chamber to mix with the high-pressure air from the compressor. The performance, efficiency, and stability of the entire engine depend on how well the injection system performs its functions.

5) Exhaust nozzle



Its sole purpose is to take the high-temperature, high-pressure gas exiting the liner and accelerate it to the highest possible velocity as it exits the engine. This high-speed exhaust jet produces the engine's thrust.

6) Impeller



It provides the highly pressurized air that is essential for powerful and efficient combustion, and it dictates the total mass of air that can flow through the engine, which is a key factor in determining the engine's overall thrust.

8.1 Joints and assembly adjustments:

Gasket:



This is a gasket used to join compressor casing and combustion chamber casing to prevent the leakage of compressed air from compressor to combustion chamber.

The development of micro jet engine is divided into three parts. They are:

- Assembly of metal combustion chamber
- Metal combustion chamber testing
- Iteration and optimization of complete system

1) Assembly of metal combustion chamber:

We have started fabrication of individual parts that are required for combustion chamber. Outer liner, outer casing, fuel injection manifold, nozzle and other small parts have to be done.

The final position of the spark plug has to be decided to ensure efficient ignition.

2) Metal combustion chamber testing:

1. After completion of this, a test has to be done with pressure gun with an adapter to deliver compressed air to the combustor for monitoring proper fabrication of combustion chamber.
2. If above test is fine, then we can test using fuel injection and ignition system using pressure gun and adapter assembly.
3. A series of tests will be conducting by varying fuel flow rates until to attain stable and sustaining combustion inside the chamber.

3) Iteration and optimization of complete system

1. Integrate the metal combustion chamber with the centrifugal compressor at inlet and nozzle at the outlet.
2. Now test it and calculate thrust output, fuel consumption rate, exhaust gas temperature.
3. With the collected data analyse with cycle analysis, find the optimum mass flow rate for surge and self-sustaining conditions and optimize the design for further iterations.

Fabrication of Inner Liner

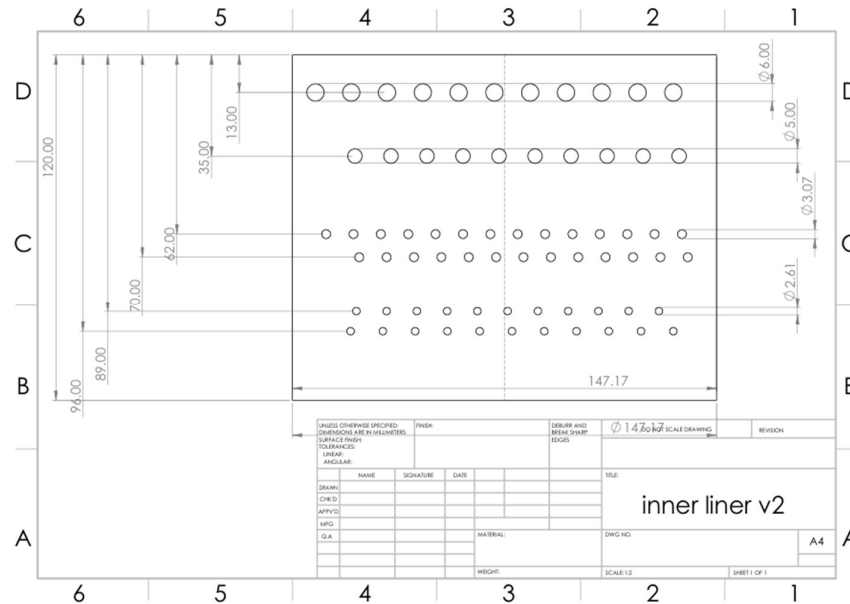
The inner liner of the combustion chamber was fabricated using a cylindrical metal tube, as per the CAD design and calculated hole pattern. The geometry and dimensions of the liner were determined from the combustor design analysis to ensure proper air-fuel mixing and cooling effectiveness.

The initial cutting of the cylindrical section was carried out using a plasma cutter machine, which provided precise cutting of the metal sheet into the required dimensions. Once the cylindrical blank was prepared, the air admission and dilution holes were created on the

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surface according to the predetermined design values. These holes were drilled using appropriately sized drill bits, ensuring that the diameter and spacing matched the calculated pattern. The hole configuration was based on design considerations for primary, secondary, and dilution air zones within the combustor.

The completed inner liner contains a carefully distributed array of holes, each dimensioned and positioned to satisfy the combustion and cooling requirements of the chamber. The fabrication process ensured dimensional accuracy while maintaining the structural integrity of the liner.



8.3 CONCLUSION:

This report demonstrates the comprehensive design and preliminary fabrication of a hybrid electric-combustion micro-jet engine, where the conventional turbine is replaced by an electric motor driving the compressor to maximize thrust through complete exhaust gas expansion through the nozzle. The theoretical foundation has been established through detailed Brayton cycle analysis and thermodynamic performance calculations, providing closed-form expressions for critical parameters including exit velocity, specific thrust, fuel-air ratio, and thermal efficiency. The iterative design process has yielded optimized component geometries for the centrifugal compressor, annular combustion chamber with precise perforation patterns for air admission and cooling, fuel injection system with eleven injector tubes, and converging-diverging exhaust nozzle. Manufacturing progress includes successful 3D printing of prototype components for initial testing, followed by metal fabrication of key parts including the inner liner with calculated hole patterns, outer casing assemblies, and fuel injection manifolds. Preliminary testing of the 3D-printed prototype has provided valuable insights into airflow characteristics, sealing requirements, and component integration challenges, informing design modifications such as reduced impeller clearances and improved gasket configurations. The experimental setup for compressor performance characterization using venturi meter mass flow measurement and pressure ratio analysis has been completed, enabling comprehensive performance mapping across various RPM ranges. Critical combustion stability parameters have been identified including Damköhler number ($Da \gtrsim 1$), equivalence ratio ($\phi \approx 0.8-1.2$), residence time constraints, and swirl number requirements ($S \gtrsim 0.6$) for stable flame anchoring in the annular combustor. The project is positioned for the next phase experimental mass flow rate estimation and optimization, involving metal combustion chamber testing, system integration, and iterative optimization to achieve target thrust performance while maintaining compressor operation above surge limits and establishing self-sustaining combustion conditions.

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